CSE203B Convex Optimization Homework 4

Ethan Chen - A69043007
Gayathri Rajesh - A69040613
Tejal Kulkarni - A69043100
Tomasz Frelek - A69042552

February 17, 2026

Contributions:

Ethan did 4.11, 4.19, 4.38, and II.2.
Gayathri did 4.1, 5.4, and II.1.
Tomasz did 5.5, 5.7, 5.8, and II.3 part i.
Tejal did 4.8, 5.11, II.3 part ii and iii.

1 Exercises from Textbook

4.1

$$\begin{aligned} \min_{x_1,x_2} \quad & f_o(x_1,x_2) \\ \text{subject to} \quad & 2x_1 + x_2 \geq 1, \\ & x_1 + 3x_2 \geq 1, \\ & x_1 \geq 0, \\ & x_2 \geq 0. \end{aligned}$$

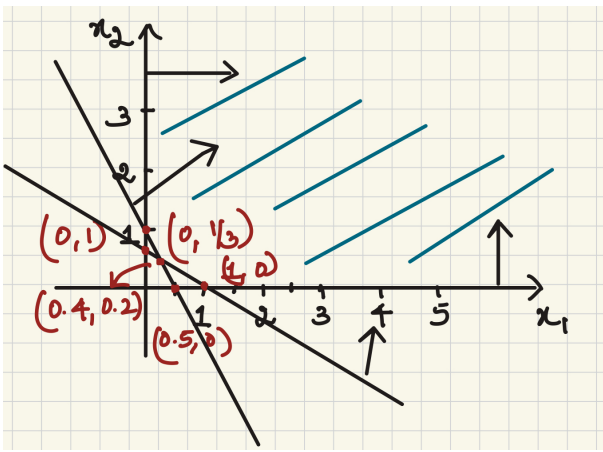

Figure 1: Sketch of the feasible region (shown in blue lines)

The feasible convex set is the region in the first quadrant given by the following points $(0,1)$, $(1,0)$ and the point of intersection of the two constraints $2x_1 + x_2 \geq 1$ and $x_1 + 3x_2 \geq 1$ given by $(0.4, 0.2)$.

(a)

$f_o(x_1, x_2) = x_1 + x_2$

$x_1 + x_2 = c$ represents the family of straight lines.

As you decrease c, the line moves towards the origin. We evaluate this function at the boundary points of the feasible set

$$f_o(0, 1) = 1$$
$$f_o(1, 0) = 1$$
$$f_o(0.4, 0.2) = 0.6$$

- **Optimal Set:** $X_{opt} = \{(0.4, 0.2)\}$
- **Optimal Value:** $p^* = 0.6$

(b)

$f_o(x_1, x_2) = -x_1 - x_2$

$$-x_1 - x_2 = c$$

also represents the family of straight lines. This can be rewritten as

$$x_1 + x_2 = -c$$

As you decrease c, the line moves away from the origin. As $x_1 \to \infty$ and $x_2 \to \infty$, the function $f_o(x_1, x_2)$ reaches its minimum value of $-\infty$

- **Optimal Set:** $X_{opt} = \{\phi\}$
- **Optimal Value:** $p^* = -\infty$

(c)

$f_o(x_1, x_2) = x_1$

$$x_1 = c$$

represents a straight line parallel to the y-axis. As c increases, this line moves further away from the origin. The minimum feasible value of $x_1 = 0$.

$$f_o(0, 1) = 0$$
$$f_o(1, 0) = 1$$
$$f_o(0.4, 0.2) = 0.4$$

- **Optimal Set:** $X_{opt} = \{(0, x_2) | x_2 \geq 1\}$
- **Optimal Value:** $p^* = 0$

(d)

$\max\{x_1, x_2\}$
Say, $\max\{x_1, x_2\} \leq \gamma$, this means $x_1 \leq \gamma$ and $x_2 \leq \gamma$
Evaluating this function f_o at the boundary points of the feasible set, we get

$$f_o(0,1) = 1$$

$$f_o(1,0) = 1$$

$$f_o(0.4, 0.2) = 0.4$$

Checking the points that lie along the line $2x_1 + x_2 = 1$
Evaluating $\max\{x_1, 1 - 2x_1\}$
After $(0.4, 0.2)$, the value of x_1 increases so the maximum value also increases.

Checking the points that lie along the line $x_1 + 3x_2 = 1$
Evaluating $\max\{1 - 3x_2, x_2\}$
Before $(0.4, 0.2)$, the value of x_2 increases so the maximum value also increases.

- **Optimal Set:** $X_{opt} = \{(0.4, 0.2)\}$
- **Optimal Value:** $p^* = 0.4$

(e)

$f_o(x_1, x_2) = x_1^2 + 9x_2^2$
Setting $x_1^2 + 9x_2^2 = c$ we get an ellipsoid.

We evaluate the ellipsoid at each of the boundary points of the feasible set:

$$f_o(0,1) = 9$$

$$f_o(1,0) = 1$$

$$f_o(0.4, 0.2) = 0.52$$

Additionally, we also check if any of the constraints are tangent to the given ellipsoid. From the constraint equation:

$$x_1 = 1 - 3x_2$$

Substituting this into the objective function f_0:

$$f(x_2) = (1 - 3x_2)^2 + 9x_2^2$$

Expanding the quadratic term:

$$f(x_2) = (1 - 6x_2 + 9x_2^2) + 9x_2^2$$

$$f(x_2) = 18x_2^2 - 6x_2 + 1$$

To find the minimum, we take the first derivative with respect to x_2 and set it to zero:

$$f'(x_2) = 36x_2 - 6 = 0$$

$$36x_2 = 6 \implies x_2 = \frac{1}{6}$$

Now, solving for x_1 using the constraint:

$$x_1 = 1 - 3\left(\frac{1}{6}\right) = 1 - \frac{1}{2} = \frac{1}{2}$$

The optimal value p^* is:

$$p^* = f_0\left(\frac{1}{2}, \frac{1}{6}\right) = \left(\frac{1}{2}\right)^2 + 9\left(\frac{1}{6}\right)^2$$

$$p^* = \frac{1}{4} + 9\left(\frac{1}{36}\right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Since the point $(\frac{1}{2}, \frac{1}{6})$ satisfies the other constraints ($2x_1 + x_2 \geq 1$ and $x_1, x_2 \geq 0$), it is the optimal solution.

- **Optimal Set:** $X_{opt} = \left\{\left(\frac{1}{2}, \frac{1}{6}\right)\right\}$
- **Optimal Value:** $p^* = 0.5$

4.8

(a) Minimizing a linear function over an affine set.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b. \end{array}$$

Let $\mathcal{R}(A)$ denote the range (column space) of A.

- **Infeasible:** If $b \notin \mathcal{R}(A)$, then the constraint set is empty and the optimal value is

$$p^\star = +\infty.$$

- **Feasible and objective constant on the affine set:** If $b \in \mathcal{R}(A)$ and $c \in \mathcal{R}(A^T)$, i.e., there exists λ with

$$c = A^T\lambda,$$

then for any feasible x,

$$c^T x = \lambda^T A x = \lambda^T b,$$

so *every feasible x is optimal* and

$$p^\star = \lambda^T b.$$

- **Feasible but unbounded below:** If $b \in \mathcal{R}(A)$ and $c \notin \mathcal{R}(A^T)$, then c has a nonzero component in $\mathcal{N}(A)$. Pick $\hat{c} \neq 0$ such that $A\hat{c} = 0$ and $c^T\hat{c} > 0$. For any feasible x_0, $x(t) = x_0 - t\hat{c}$ satisfies $Ax(t) = b$ and

$$c^T x(t) = c^T x_0 - t\, c^T\hat{c} \to -\infty \quad (t \to \infty),$$

hence

$$p^\star = -\infty.$$

$$p^\star = \begin{cases} +\infty, & b \notin \mathcal{R}(A), \\ \lambda^T b, & c = A^T\lambda \text{ for some } \lambda \text{ and } b \in \mathcal{R}(A), \\ -\infty, & \text{otherwise.} \end{cases}$$

(b) Minimizing a linear function over a halfspace.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & a^T x \le b, \end{array} \qquad a \neq 0.$$

This problem is always feasible. Decompose c into a component parallel to a and a component orthogonal to a:

$$c = \lambda a + \hat{c}, \qquad a^T \hat{c} = 0.$$

- **Unbounded if $\lambda > 0$:** Take $x = -ta$. Then $a^T x = -t\|a\|_2^2 \le b$ for large t, and

$$c^T x = (\lambda a + \hat{c})^T(-ta) = -t\lambda\|a\|_2^2 \to -\infty.$$

- **Unbounded if $\hat{c} \neq 0$:** Take $x = \frac{b}{\|a\|_2^2} a - t\hat{c}$. Then $a^T x = b$ (since $a^T \hat{c} = 0$) and

$$c^T x = \lambda b - t\|\hat{c}\|_2^2 \to -\infty.$$

- **Bounded case:** If $\hat{c} = 0$ and $\lambda \le 0$ (i.e., $c = \lambda a$ with $\lambda \le 0$), then minimizing $\lambda a^T x$ with $\lambda \le 0$ forces $a^T x$ to be as large as possible under $a^T x \le b$, hence $a^T x = b$ at optimum and

$$p^\star = \lambda b.$$

  One optimal point is $x^\star = \frac{b}{\|a\|_2^2} a$.

$$p^\star = \begin{cases} \lambda b, & c = \lambda a \text{ for some } \lambda \le 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

(c) Minimizing a linear function over a rectangle (box).

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \ell \preceq x \preceq u, \end{array} \qquad \ell \preceq u.$$

The objective and constraints are separable across coordinates. For each i,

$$\min_{\ell_i \le x_i \le u_i} c_i x_i = \begin{cases} c_i \ell_i, & c_i > 0, \\ c_i u_i, & c_i < 0, \\ \text{any value}, & c_i = 0. \end{cases}$$

Thus an optimal solution is

$$x_i^\star = \begin{cases} \ell_i, & c_i > 0, \\ u_i, & c_i < 0, \\ \text{any in } [\ell_i, u_i], & c_i = 0, \end{cases} \qquad i = 1, \ldots, n.$$

(d) Minimizing a linear function over the probability simplex.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \mathbf{1}^T x = 1, \quad x \succeq 0. \end{array}$$

Let $c_{\min} = \min_i c_i$ and choose any index $i^\star \in \arg\min_i c_i$. Since $c^T x \geq c_{\min} \mathbf{1}^T x = c_{\min}$, the minimum is achieved by putting all mass on a minimizer:

$$x^\star = e_{i^\star}, \qquad p^\star = c_{\min}.$$

If the equality is replaced by $\mathbf{1}^T x \leq 1$ (with $x \succeq 0$), we may also choose $x = 0$. Hence

$$p^\star = \min\{0, c_{\min}\}.$$

(e) Minimizing over a unit box with a total budget constraint.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \mathbf{1}^T x = \alpha, \quad 0 \preceq x \preceq \mathbf{1}, \end{array} \qquad 0 \leq \alpha \leq n.$$

Sort the coefficients so that

$$c_1 \leq c_2 \leq \cdots \leq c_n.$$

If α is an integer: An optimal solution is

$$x_1^\star = \cdots = x_\alpha^\star = 1, \qquad x_{\alpha+1}^\star = \cdots = x_n^\star = 0,$$

and

$$p^\star = \sum_{i=1}^{\alpha} c_i.$$

(If there are ties at c_α, any distribution among tied indices is optimal.)

If α is not an integer: Let $k = \lfloor \alpha \rfloor$ and $\delta = \alpha - k \in (0, 1)$. Then an optimal solution is

$$x_1^\star = \cdots = x_k^\star = 1, \qquad x_{k+1}^\star = \delta, \qquad x_{k+2}^\star = \cdots = x_n^\star = 0,$$

and

$$p^\star = \sum_{i=1}^{k} c_i + \delta\, c_{k+1}.$$

If the equality is replaced by $\mathbf{1}^T x \leq \alpha$ (and $0 \preceq x \preceq \mathbf{1}$): you only buy variables with negative cost first. The optimal value is the sum of the α smallest *nonpositive* coefficients (if α is integer; with fractional α, the last one can be fractional similarly).

(f) Minimizing over a unit box with a weighted budget constraint.

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & d^T x = \alpha, \quad 0 \preceq x \preceq \mathbf{1}, \end{array} \qquad d \succ 0, \quad 0 \leq \alpha \leq \mathbf{1}^T d.$$

Change variables $y_i = d_i x_i$ so that $0 \leq y_i \leq d_i$ and $\mathbf{1}^T y = \alpha$:

$$\min \sum_{i=1}^{n} \frac{c_i}{d_i} y_i \quad \text{s.t.} \quad \mathbf{1}^T y = \alpha, \quad 0 \preceq y \preceq d.$$

Sort ratios:

$$\frac{c_1}{d_1} \le \frac{c_2}{d_2} \le \cdots \le \frac{c_n}{d_n}.$$

Let

$$k = \max\left\{ i \in \{0, 1, \ldots, n\} : \sum_{j=1}^{i} d_j \le \alpha \right\} \quad (\text{with } \sum_{j=1}^{0} d_j = 0),$$

and define the remainder

$$r = \alpha - \sum_{j=1}^{k} d_j, \qquad 0 \le r \le d_{k+1} \ \ (\text{if } k < n).$$

Then an optimal solution in y is

$$y_1^\star = d_1, \ \ldots, \ y_k^\star = d_k, \qquad y_{k+1}^\star = r, \qquad y_{k+2}^\star = \cdots = y_n^\star = 0,$$

and in the original variables

$$x_1^\star = \cdots = x_k^\star = 1, \qquad x_{k+1}^\star = \frac{r}{d_{k+1}} = \frac{\alpha - \sum_{j=1}^{k} d_j}{d_{k+1}}, \qquad x_{k+2}^\star = \cdots = x_n^\star = 0.$$

$$p^\star = \sum_{i=1}^{n} c_i x_i^\star = \sum_{i=1}^{k} c_i \cdot 1 \ + \ c_{k+1} \cdot \frac{r}{d_{k+1}} = \sum_{i=1}^{k} c_i \ + \ \frac{c_{k+1}}{d_{k+1}} r, \qquad (k < n).$$

4.11

(a) Minimize $||Ax - b||_\infty$

Equivalent LP

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{s.t.} \quad & -t\mathbf{1} \le Ax - b \le t\mathbf{1} \end{aligned}$$

Given any x, the minimal feasible $t = ||Ax - b||_\infty$. At the optimum (x^*, t^*) of the LP, we must have $t^* = ||Ax^* - b||_\infty$.

(b) Minimize $||Ax - b||_1$

Equivalent LP

$$\begin{aligned} \min_{x,u} \quad & \mathbf{1}^\top u \\ \text{s.t.} \quad & -u \le Ax - b \le u, \\ & u \ge 0 \end{aligned}$$

For any fixed x, the minimal feasible u is $u_i = |(Ax - b)_i|$ so $\mathbf{1}^\top u = ||Ax - b||_1$. At optimum (x^*, u^*), we must have $u^* = |Ax^* - b|$.

(c) Minimize $||Ax - b||_1 \text{s.t.} ||x||_\infty \le 1$

Equivalent LP

$$\begin{aligned} \min_{x,u} \quad & \mathbf{1}^\top u \\ \text{s.t.} \quad & -u \le Ax - b \le u, \\ & u \ge 0, \\ & -\mathbf{1} \le x \le \mathbf{1} \end{aligned}$$

The relation is the same as part (b). Feasiblity matches exactly via the last constraint, $-1 \leq x \leq 1$.

(d) Minimize $||x||_1$s.t.$||Ax - b||_\infty \leq 1$

Equivalent LP

$$\begin{aligned} \min_{x,v} \quad & \mathbf{1}^\top v \\ \text{s.t.} \quad & -v \leq x \leq v, \\ & v \geq 0, \\ & -\mathbf{1} \leq Ax - b \leq \mathbf{1} \end{aligned}$$

For any feasible x, the minimal v is $v_i = |x_i|$ so the objective is equal to $||x||_1$. Any feasible (x, v) in the LP has $||Ax - b||_\infty \leq 1$ and $||x||_1 \leq \mathbf{1}^\top v$ at the optimum $v = |x|$.

(e) Minimize $||Ax - b||_1 + ||x||_\infty$

$$\begin{aligned} \min_{x,u,t} \quad & \mathbf{1}^\top u + t \\ \text{s.t.} \quad & -u \leq Ax - b \leq u, \\ & u \geq 0, \\ & -t\mathbf{1} \leq x \leq t\mathbf{1} \\ & t \geq 0 \end{aligned}$$

For a fixed x, the minimal u is $|Ax - b|$ and the minimal t is $||x||_\infty$. Our LP objective becomes the same as the given expression: $||Ax - b||_1 + ||x||_\infty$.

At the optimum, $u^* = |Ax^* - b|$ and $t^* = ||x^*||_\infty$.

4.19

(a) Show that this is a quasiconvex optimization problem.

$$c^\top x + d \geq d - ||c||_1 ||x||_\infty \geq d - ||c||_1 > 0$$

by Holder's inequality and the assumption that $d - ||c||_1 > 0$ so $c^\top x + d > 0$.

So the objective function has a positive denominator. We need to show that the sublevel sets are convex.

Fix $\alpha \in \mathbb{R}$. Define

$$S_\alpha = \left\{ x : ||x||_\infty \leq 1, \frac{||Ax - b||_1}{c^\top x + d} \leq \alpha \right\}$$

Since $c^\top x + d > 0$ for all feasible x, we can use this to get

$$||Ax - b||_1 \leq \alpha(c^\top x + d)$$

Case 1 $\alpha < 0$: LHS ≥ 0 and right side < 0 so $S_\alpha =$. Empty set is (vacuously convex).

Case 2 $\alpha \geq 0$:

$$||Ax - b||_1 - \alpha c^\top x \leq \alpha d$$

The first term on LHS is convex, and the second term is linear so LHS is convex. The sublevel set of a convex function is convex.

The intersection of two convex sets, case 2's inequality set and $||x||_\infty \leq 1$, is convex.
So every S_α is convex, thus f is **quasiconvex** on $||x||_\infty \leq 1$.

(b) Show that it is equivalent to ...
We can let

$$t = \frac{1}{c^\top x + d}$$

Then do

$$y = tx = \frac{x}{c^\top x + d}$$

$$\frac{||A(y/t) - b||_1}{c^\top (y/t) + d} = \frac{\frac{1}{t}||Ay - bt||_1}{\frac{1}{t}(c^\top y + dt)} = \frac{||Ay - bt||_1}{c^\top y + dt}$$

$$c^\top y + dt = c^\top (tx) + dt = t(c^\top x + d) = 1$$

The original constraint is $||x||_\infty \leq 1$. We can replace x with y/t to get

$$||y/t||_\infty \leq 1 \implies \frac{1}{t}||y||_\infty \leq 1 \implies ||y||_\infty \leq t$$

Since $c^\top x + d > 0$ and $t = 1/(c^\top x + d)$, we know t must be > 0.
Conversely, given any feasible (y, t) with $t > 0$, we can define $x = y/t$.
Then, since $\|y\|_\infty \leq t$ and $c^\top x + d = (c^\top y + dt)/t = 1/t$, which yields $t = 1/(c^\top x + d)$,

$$||x||_\infty \leq 1$$

Then, we see that $||Ay - bt||_1 = t||Ax - b||_1$, so the objective values match.
This gives us the convex optimization problem stated in the question part b.

4.38

(a) Show that the generalized ...
We can first define

$$D_a := \text{diag}(a) = R^\top AR, \quad D_b := \text{diag}(b) = R^\top BR$$

Then by multipling the left by $R^{-\top}$ and the right by R^{-1}, we get

$$A = R^{-\top} D_a R^{-1}, \quad B = R^{-\top} D_b R^{-1}$$

$$\begin{aligned} \det(\lambda B - A) &= \det(\lambda R^{-\top} D_b R^{-1} - R^{-\top} D_a R^{-1}) \\ &= \det(R^{-\top}) \det(\lambda D_b - D_a) \det(R^{-1}) \end{aligned}$$

Because R is nonsingular, both $\det(R^{-1}) \neq 0$ and $\det(R^{-\top}) \neq 0$, so

$$\det(\lambda B - A) = 0 \iff \det(\lambda D_b - D_a) = 0$$

But we're given $\lambda D_b - D_a = \text{diag}(\lambda b_i - a_i)$, so

$$\det(\lambda D_b - D_a) = \prod_{i=1}^{n} (\lambda b_i - a_i)$$

So the roots are exactly $\lambda_i = \frac{a_i}{b_i}, \quad i = 1, \ldots, n$. These are real b/c $a_i, b_i \in \mathbb{R}$ and $b_i \neq 0$.

(b) Express the solution of the SDP ...

We can rewrite $tB \preceq A$ as $R^\top(A - tB)R \succeq 0$ since R is nonsingular.

$$\operatorname{diag}(a) - t\operatorname{diag}(b) \succeq 0 \implies a_i - tb_i \geq 0 \text{ for } i = 1, \ldots, n$$

This gives $a_i \geq tb_i \quad \forall i$.

Feasible t satisfy

$$\max_{b_i<0} \frac{a_i}{b_i} \leq t \leq \min_{b_i>0} \frac{a_i}{b_i}$$

Case 1: $c > 0$ so we need to minimize t, s.t. $tb_i \leq a_i$. If $b_i > 0$, $t \leq a_i/b_i$. And if $b_i < 0$, $t \geq a_i/b_i$. The minimum is $t^* = \max\{a_i/b_i \mid b_i < 0\}$.

Case 2: $c < 0$ so we need to maximize t; similarly, we can find

$$t^* = \min\{a_i/b_i \mid b_i > 0\}$$

5.4

Consider the primal Linear Program (LP) in inequality form:

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b \end{array}$$

where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Let $w \in \mathbb{R}^m_+$ be a vector of weights. For any feasible x, the inequality $w^T Ax \leq w^T b$ holds.

(a) Minimum value of $c^T x$ over the halfspace H_w

We seek to minimize the objective over the relaxed halfspace $H_w = \{x \mid (A^T w)^T x \leq w^T b\}$. The problem is:

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & (A^T w)^T x \leq w^T b \end{array}$$

A linear function $c^T x$ is bounded below on a halfspace only if the c can be expressed as $\lambda A^T w$. Specifically, for the minimum to be finite, we require:

$$c = -A^T w \quad \text{for some } w \succeq 0$$

If this condition holds, the minimum occurs on the boundary $(A^T w)^T x = w^T b$. Substituting $A^T w = -c$, we get:

$$(-c)^T x = w^T b \implies c^T x = -b^T w$$

Otherwise, if c cannot be expressed as $-A^T w$, the minimum value is $-\infty$.

(b) The Best Lower Bound

To find the best lower bound, we maximize the value obtained in part (a) over $w \succeq 0$:

$$\begin{array}{ll} \text{maximize} & -b^T w \\ \text{subject to} & A^T w + c = 0 \\ & w \succeq 0 \end{array}$$

(c) Relation to the Lagrange Dual

The Lagrangian for the primal LP is:

$$L(x, w) = c^T x + w^T(Ax - b) = (c + A^T w)^T x - b^T w$$

The Lagrange dual function $g(w)$ is defined as $\inf_x L(x, w)$:

$$g(w) = \begin{cases} -b^T w & \text{if } A^T w + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

Maximizing $g(w)$ for $w \succeq 0$ yields exactly the same optimization problem derived via the geometric relaxation in part (b). This confirms that the dual variable w_i can be interpreted as the weight assigned to the i-th constraint to form a supporting halfspace for the feasible set.

5.5

We introduce Lagrange multiplier vectors $\lambda \succeq 0$ for the inequality constraint $Gx - h \leq 0$, and ν for the equality constraint $Ax - b = 0$.

The Lagrangian is the sum of the objective function and the constraints weighted by their multipliers:

$$L(x, \lambda, \nu) = c^T x + \lambda^T(Gx - h) + \nu^T(Ax - b)$$

Rearranging the terms to isolate the primal variable x:

$$L(x, \lambda, \nu) = (c^T + \lambda^T G + \nu^T A)x - \lambda^T h - \nu^T b$$

The dual function is defined as $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$.

Because this function is affine with respect to x, it is only bounded below if the coefficient multiplying x is exactly the zero vector. If it is not zero, x can be chosen to make the function approach $-\infty$.

Therefore, for the infimum to be finite, we require:

$$c^T + \lambda^T G + \nu^T A = 0$$

Taking the transpose, this is equivalent to:

$$G^T \lambda + A^T \nu + c = 0$$

If this condition is met, the x term zeroes out, leaving only the constant terms. This gives us the dual function:

$$g(\lambda, \nu) = \begin{cases} -h^T \lambda - b^T \nu & \text{if } G^T \lambda + A^T \nu + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

Then the dual problem can be stated as:

$$\begin{aligned} \text{maximize} \quad & -h^T \lambda - b^T \nu \\ \text{subject to} \quad & G^T \lambda + A^T \nu + c = 0 \\ & \lambda \succeq 0 \end{aligned}$$

5.7

a

We introduce a Lagrange multiplier vector $\nu \in \mathbf{R}^m$ for the equality constraints $a_i^T x + b_i - y_i = 0$. The Lagrangian $L(x, y, \nu)$ is:

$$L(x, y, \nu) = \max_{i=1,\ldots,m} y_i + \sum_{i=1}^{m} \nu_i (a_i^T x + b_i - y_i)$$

Grouping the terms by the primal variables x and y, we get:

$$L(x, y, \nu) = \left(\sum_{i=1}^{m} \nu_i a_i \right)^T x + \left(\max_{i=1,\ldots,m} y_i - \sum_{i=1}^{m} \nu_i y_i \right) + \sum_{i=1}^{m} \nu_i b_i$$

The dual function is $g(\nu) = \inf_{x,y} L(x, y, \nu)$. We minimize over x and y separately.

- **Minimizing over x:** The term involving x is linear. For the infimum to be bounded below, the coefficient of x must be exactly zero:

$$\sum_{i=1}^{m} \nu_i a_i = 0$$

- **Minimizing over y:** We must find $\inf_y \left(\max_i y_i - \nu^T y \right)$. If any $\nu_k < 0$, we can let $y_k \to -\infty$ while keeping other $y_j = 0$, driving the expression to $-\infty$. Thus, we require $\nu \succeq 0$.

  If we shift all y_i by a constant c, the expression becomes $\max_i y_i - \nu^T y + c(1 - \sum_{i=1}^m \nu_i)$. If $\sum_{i=1}^m \nu_i \neq 1$, we can choose $c \to \pm\infty$ to drive the expression to $-\infty$. Thus, we require $\sum_{i=1}^m \nu_i = 1$.

  If both $\nu \succeq 0$ and $\sum_{i=1}^m \nu_i = 1$ hold, $\nu^T y$ is a convex combination of y, meaning $\nu^T y \leq \max_j y_j$. Therefore, the minimum value of $\max_j y_j - \nu^T y$ is 0.

If these conditions are met, the x and y terms vanish, leaving:

$$g(\nu) = \begin{cases} \sum_{i=1}^m \nu_i b_i & \text{if } \sum_{i=1}^m \nu_i a_i = 0,\ \sum_{i=1}^m \nu_i = 1,\ \nu \succeq 0 \\ -\infty & \text{otherwise} \end{cases}$$

Then the dual problem is:

$$\begin{array}{ll} \text{maximize} & \displaystyle\sum_{i=1}^{m} b_i \nu_i \\ \text{subject to} & \displaystyle\sum_{i=1}^{m} \nu_i a_i = 0 \\ & \displaystyle\sum_{i=1}^{m} \nu_i = 1 \\ & \nu \succeq 0 \end{array}$$

b

We cast the minimax problem into an LP by introducing an epigraph variable $t \in \mathbf{R}$:

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & a_i^T x + b_i \leq t, \quad i = 1, \ldots, m \end{array}$$

Introduce a multiplier vector $\lambda \in \mathbf{R}^m$ for the inequality constraints $a_i^T x + b_i - t \leq 0$, with $\lambda \succeq 0$. The Lagrangian is:

$$L(x, t, \lambda) = t + \sum_{i=1}^{m} \lambda_i (a_i^T x + b_i - t)$$

Rearranging to isolate x and t:

$$L(x, t, \lambda) = \left(1 - \sum_{i=1}^{m} \lambda_i\right) t + \left(\sum_{i=1}^{m} \lambda_i a_i\right)^T x + \sum_{i=1}^{m} \lambda_i b_i$$

To find the dual function $g(\lambda) = \inf_{x,t} L(x, t, \lambda)$, the coefficients of the unconstrained variables x and t must be zero:

- Coefficient of t: $1 - \sum_{i=1}^{m} \lambda_i = 0 \implies \sum_{i=1}^{m} \lambda_i = 1$
- Coefficient of x: $\sum_{i=1}^{m} \lambda_i a_i = 0$

If these hold, $g(\lambda) = \sum_{i=1}^{m} \lambda_i b_i$. This yields the LP dual problem:

$$\begin{array}{ll} \text{maximize} & \displaystyle\sum_{i=1}^{m} b_i \lambda_i \\ \text{subject to} & \displaystyle\sum_{i=1}^{m} \lambda_i a_i = 0 \\ & \displaystyle\sum_{i=1}^{m} \lambda_i = 1 \\ & \lambda \succeq 0 \end{array}$$

Note that this new dual is equivalent to the dual derived in part a, where v = λ

c

Since all terms in the sum $\sum_{i=1}^{m} \exp(y_i(x))$ are strictly positive, the total sum is strictly greater than or equal to any individual term, including the maximum term:

$$\exp\left(\max_i y_i(x)\right) \leq \sum_{i=1}^{m} \exp(y_i(x))$$

Because the natural logarithm is a strictly monotonically increasing function, we can take the log of both sides without changing the direction of the inequality:

$$\max_i y_i(x) \leq \log\left(\sum_{i=1}^{m} \exp(y_i(x))\right)$$

This inequality holds for all x. Taking the infimum over x on both sides yields $p^\star_{\text{pwl}} \leq p^\star_{\text{gp}}$, which rearranges to:

$$0 \leq p^\star_{\text{gp}} - p^\star_{\text{pwl}}$$

We can also bound the sum from above. Every term in the sum is less than or equal to the maximum term. Therefore, replacing every term in the sum with the maximum term provides an upper bound:

$$\sum_{i=1}^{m} \exp(y_i(x)) \leq \sum_{i=1}^{m} \exp\left(\max_j y_j(x)\right) = m \exp\left(\max_i y_i(x)\right)$$

Taking the logarithm of both sides gives:

$$\log\left(\sum_{i=1}^{m} \exp(y_i(x))\right) \leq \log\left(m \exp\left(\max_i y_i(x)\right)\right)$$

Using the logarithm product rule $\log(ab) = \log(a) + \log(b)$:

$$\log\left(\sum_{i=1}^{m} \exp(y_i(x))\right) \leq \log m + \max_i y_i(x)$$

Taking the infimum over x yields $p^\star_{\text{gp}} \leq p^\star_{\text{pwl}} + \log m$, which rearranges to:

$$p^\star_{\text{gp}} - p^\star_{\text{pwl}} \leq \log m$$

Combining the lower and upper bounds gives the result:

$$0 \leq p^\star_{\text{gp}} - p^\star_{\text{pwl}} \leq \log m$$

d

The sum is greater than or equal to its maximum term:

$$\exp\left(\gamma \max_i y_i(x)\right) \leq \sum_{i=1}^{m} \exp(\gamma y_i(x))$$

Taking the logarithm of both sides:

$$\gamma \max_i y_i(x) \leq \log\left(\sum_{i=1}^{m} \exp(\gamma y_i(x))\right)$$

Since $\gamma > 0$, dividing by γ preserves the inequality:

$$\max_i y_i(x) \leq \frac{1}{\gamma} \log\left(\sum_{i=1}^{m} \exp(\gamma y_i(x))\right)$$

Taking the infimum over x yields $p^\star_{\text{pwl}} \leq p^\star_\gamma$, or:

$$0 \leq p^\star_\gamma - p^\star_{\text{pwl}}$$

Replacing all terms in the sum with the maximum term:

$$\sum_{i=1}^{m} \exp(\gamma y_i(x)) \leq m \exp\left(\gamma \max_i y_i(x)\right)$$

Taking the logarithm:

$$\log\left(\sum_{i=1}^{m}\exp(\gamma y_i(x))\right) \leq \log m + \gamma \max_i y_i(x)$$

Dividing by $\gamma > 0$:

$$\frac{1}{\gamma}\log\left(\sum_{i=1}^{m}\exp(\gamma y_i(x))\right) \leq \frac{\log m}{\gamma} + \max_i y_i(x)$$

Taking the infimum over x yields $p^\star_\gamma \leq p^\star_{\text{pwl}} + \frac{\log m}{\gamma}$, or:

$$p^\star_\gamma - p^\star_{\text{pwl}} \leq \frac{\log m}{\gamma}$$

Combining the bounds gives:

$$0 \leq p^\star_\gamma - p^\star_{\text{pwl}} \leq \frac{\log m}{\gamma}$$

As the parameter $\gamma \to \infty$, the upper bound term $\frac{\log m}{\gamma} \to 0$. Because the error is bounded below by 0, the difference $p^\star_\gamma - p^\star_{\text{pwl}}$ must also approach 0. Therefore, $\lim_{\gamma\to\infty} p^\star_\gamma = p^\star_{\text{pwl}}$. The smooth approximation becomes exact as γ increases.

5.8

The first dual problem is formulated as:

$$\begin{array}{ll} \text{maximize} & -b^T\nu - \lambda_1^T u + \lambda_2^T l \\ \text{subject to} & A^T\nu + c = \lambda_2 - \lambda_1 \\ & \lambda_1 \succeq 0, \quad \lambda_2 \succeq 0 \end{array}$$

Let us define a new vector $y = A^T\nu + c$ to simplify the notation. The equality constraint then becomes $y = \lambda_2 - \lambda_1$.

We substitute $\lambda_2 = y + \lambda_1$ into the objective function:

$$-b^T\nu - u^T\lambda_1 + l^T(y + \lambda_1)$$

Rearranging the terms groups the λ_1 variables together:

$$-b^T\nu + l^T y - (u - l)^T\lambda_1$$

Because $l \preceq u$, the vector $(u - l)$ is strictly non-negative. Therefore, to maximize the overall expression, we must make λ_1 as small as possible.

We have two constraints on λ_1:

1. $\lambda_1 \succeq 0$
2. $\lambda_1 = \lambda_2 - y \implies \lambda_1 \succeq -y$ (since $\lambda_2 \succeq 0$)

Combining these, the smallest possible valid value for each component of λ_1 is the maximum of 0 and $-y_i$. This corresponds to the definition of the negative part of y, denoted as y^-:

$$(\lambda_1)_i = \max\{0, -y_i\} = y_i^-$$

Using the equality constraint $\lambda_2 = y + \lambda_1$, we can find the corresponding optimal value for λ_2:

$$(\lambda_2)_i = y_i + \max\{0, -y_i\} = \max\{y_i, 0\} = y_i^+$$

Therefore, the optimal choices for the multipliers are:

$$\lambda_1^\star = (A^T\nu + c)^-$$
$$\lambda_2^\star = (A^T\nu + c)^+$$

Finally, we substitute these optimal values $\lambda_1^\star$ and $\lambda_2^\star$ back into the original objective function:

$$-b^T\nu - u^T(A^T\nu + c)^- + l^T(A^T\nu + c)^+$$

Because we have analytically solved for the best possible λ_1 and λ_2 given any ν, we no longer need the constraints. We are left with an unconstrained maximization problem over the single remaining variable ν:

$$\text{maximize} \quad -b^T\nu - u^T(A^T\nu + c)^- + l^T(A^T\nu + c)^+$$

This exactly matches the other unconstrained dual problem.

Problem. Derive the dual of

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^N \|A_i x + b_i\|_2 \; + \; \frac{1}{2}\|x - x_0\|_2^2,$$

where $A_i \in \mathbb{R}^{m_i\times n}$, $b_i \in \mathbb{R}^{m_i}$, and $x_0 \in \mathbb{R}^n$.

Introduce auxiliary variables. Let $y_i \in \mathbb{R}^{m_i}$ and impose

$$y_i = A_i x + b_i, \qquad i = 1, \ldots, N.$$

Then the problem is equivalent to

$$\begin{array}{ll} \text{minimize} & \displaystyle\sum_{i=1}^N \|y_i\|_2 \; + \; \frac{1}{2}\|x - x_0\|_2^2 \\ \text{subject to} & y_i - A_i x - b_i = 0, \quad i = 1, \ldots, N. \end{array}$$

Form the Lagrangian. Introduce Lagrange multipliers $z_i \in \mathbb{R}^{m_i}$ for the equality constraints.

$$L(x, y, z) = \sum_{i=1}^N \|y_i\|_2 + \frac{1}{2}\|x - x_0\|_2^2 - \sum_{i=1}^N z_i^T\left(y_i - A_i x - b_i\right).$$

Expand the terms:

$$L(x, y, z) = \sum_{i=1}^N \left(\|y_i\|_2 - z_i^T y_i\right) + \frac{1}{2}\|x - x_0\|_2^2 + \sum_{i=1}^N z_i^T A_i x + \sum_{i=1}^N z_i^T b_i.$$

The dual function is

$$g(z) = \inf_{x,y} \; L(x, y, z) = \inf_x \left[\frac{1}{2}\|x - x_0\|_2^2 + \sum_{i=1}^N z_i^T A_i x\right] \; + \; \sum_{i=1}^N \inf_{y_i}\left[\|y_i\|_2 - z_i^T y_i\right] \; + \; \sum_{i=1}^N z_i^T b_i.$$

Minimize over y_i Consider

$$\phi_i(z_i) := \inf_{y_i} \left(\|y_i\|_2 - z_i^T y_i \right).$$

By Cauchy–Schwarz, $z_i^T y_i \le \|z_i\|_2 \|y_i\|_2$, hence

$$\|y_i\|_2 - z_i^T y_i \;\ge\; \|y_i\|_2 - \|z_i\|_2 \|y_i\|_2 = (1 - \|z_i\|_2)\|y_i\|_2.$$

- If $\|z_i\|_2 \le 1$, then $(1 - \|z_i\|_2)\|y_i\|_2 \ge 0$ for all y_i, so the infimum is attained at $y_i = 0$ and $\phi_i(z_i) = 0$.
- If $\|z_i\|_2 > 1$, choose $y_i = t z_i$ with $t \to +\infty$. Then $\|y_i\|_2 - z_i^T y_i = t\|z_i\|_2 - t\|z_i\|_2^2 = -t\|z_i\|_2(\|z_i\|_2 - 1) \to -\infty$. So $\phi_i(z_i) = -\infty$.

Therefore

$$\inf_{y_i} \left(\|y_i\|_2 - z_i^T y_i \right) = \begin{cases} 0, & \|z_i\|_2 \le 1, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence, to have a finite dual value, we must enforce $\|z_i\|_2 \le 1$ for each i.

Minimize over x Define

$$s := \sum_{i=1}^N A_i^T z_i \in \mathbb{R}^n.$$

The x-dependent part of the Lagrangian is

$$\frac{1}{2}\|x - x_0\|_2^2 + s^T x.$$

Take gradient w.r.t. x and set to zero:

$$\nabla_x \left(\frac{1}{2}\|x - x_0\|_2^2 + s^T x \right) = (x - x_0) + s = 0 \quad \Longrightarrow \quad x^\star = x_0 - s = x_0 - \sum_{i=1}^N A_i^T z_i.$$

Now evaluate the infimum value by substitution. Using $x = x_0 - s$,

$$\frac{1}{2}\|x - x_0\|_2^2 + s^T x = \frac{1}{2}\|-s\|_2^2 + s^T(x_0 - s) = \frac{1}{2}\|s\|_2^2 + s^T x_0 - \|s\|_2^2 = s^T x_0 - \frac{1}{2}\|s\|_2^2.$$

Collect terms to get the dual function. When $\|z_i\|_2 \le 1$ for all i, the y-infima contribute 0 and we have

$$g(z) = \left(s^T x_0 - \frac{1}{2}\|s\|_2^2 \right) + \sum_{i=1}^N z_i^T b_i \quad \text{where } s = \sum_{i=1}^N A_i^T z_i.$$

Since $s^T x_0 = \left(\sum_{i=1}^N A_i^T z_i \right)^T x_0 = \sum_{i=1}^N z_i^T (A_i x_0)$, we can write

$$g(z) = \sum_{i=1}^N z_i^T (A_i x_0 + b_i) - \frac{1}{2} \left\| \sum_{i=1}^N A_i^T z_i \right\|_2^2, \qquad \text{if } \|z_i\|_2 \le 1 \ \forall i,$$

and $g(z) = -\infty$ otherwise.

Dual problem. Maximize the dual function subject to the feasibility constraints:

$$\begin{array}{ll} \text{maximize} & \displaystyle\sum_{i=1}^N (A_i x_0 + b_i)^T z_i \;-\; \frac{1}{2} \left\| \sum_{i=1}^N A_i^T z_i \right\|_2^2 \\ \text{subject to} & \|z_i\|_2 \le 1, \qquad i = 1, \ldots, N. \end{array}$$

2 Additional Exercises

Solution for II.1

Given the system parameters:

$$A = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ -1 \\ -2 \\ -2 \end{bmatrix}$$

Primal Problem (P)

$$\begin{aligned} \text{minimize} \quad & f(x) = c^T x = -x_1 - x_2 - 2x_3 - 2x_4 \\ \text{subject to} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Dual Problem (D)

$$\begin{aligned} \text{maximize} \quad & g(y) = -b^T y = -(y_1 + y_2 + 2y_3 + 0y_4 + y_5) \\ \text{subject to} \quad & A^T y \geq -c \\ & y \geq 0 \end{aligned}$$

Primal Solution

Expanding the constraints $Ax \leq b$:

1. $2x_1 + x_2 + 2x_3 \leq 1$
2. $x_2 + x_4 \leq 1$
3. $x_1 \leq 2$
4. $x_1 + x_2 + x_3 + x_4 \leq 0$
5. $x_3 + x_4 \leq 1$

Given the non-negativity constraint $x \geq 0$, the fourth constraint $x_1 + x_2 + x_3 + x_4 \leq 0$ implies that each individual variable must be zero:

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 0$$

Optimal Solution: $x^* = [0, 0, 0, 0]^T$
Optimal Value: $f(x^*) = 0$

Dual Solution

The dual constraints $A^T y \geq [1, 1, 2, 2]^T$ are:

1. $2y_1 + 0y_2 + 1y_3 + 1y_4 + 0y_5 \geq 1$
2. $1y_1 + 1y_2 + 0y_3 + 1y_4 + 0y_5 \geq 1$
3. $2y_1 + 0y_2 + 0y_3 + 1y_4 + 1y_5 \geq 2$
4. $0y_1 + 1y_2 + 0y_3 + 1y_4 + 1y_5 \geq 2$

To maximize $g(y) = -(y_1 + y_2 + 2y_3 + y_5)$, we set $y_1 = y_2 = y_3 = y_5 = 0$. To satisfy constraints (3) and (4), we must have $y_4 \geq 2$. Since y_4 has a coefficient of 0 in the objective, any $y_4 \geq 2$ works.

Optimal Solution (Example): $y^* = [0, 0, 0, 2, 0]^T$

Optimal Value: $g(y^*) = 0$

Strong duality holds as $f(x^*) = g(y^*) = 0$. The primal feasible region is restricted to a single point (the origin) due to the constraint $b_4 = 0$. Minimize $f_0(x) = c^T x$ subject to $Ax = b$, where $x \in \mathbb{R}^4$. The parameters are given as:

$$A = \begin{bmatrix} 2 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ -1 \\ -2 \\ -2 \end{bmatrix}$$

Primal Problem (P)

The primal problem is to find $x \in \mathbb{R}^4$ that minimizes:

$$f_0(x) = -x_1 - x_2 - 2x_3 - 2x_4$$

subject to the linear system $Ax = b$.

Dual Problem(D)

The Lagrangian for the equality-constrained problem is:

$$L(x, \nu) = c^T x + \nu^T (Ax - b) = (c + A^T \nu)^T x - \nu^T b$$

The dual function $g(\nu) = \inf_x L(x, \nu)$ is finite only if $A^T \nu + c = 0$. Thus, the dual problem is:

$$\begin{aligned} \text{maximize} \quad & g(\nu) = -b^T \nu \\ \text{subject to} \quad & A^T \nu = -c \end{aligned}$$

Feasibility Analysis

To check the feasibility of the primal, we examine the system of equations $Ax = b$:

1. $2x_1 + x_2 + 2x_3 = 1$
2. $x_2 + x_4 = 1$
3. $x_1 = 2$
4. $x_1 + x_2 + x_3 + x_4 = 0$
5. $x_3 + x_4 = 1$

Substituting (3) into (4) gives $2 + x_2 + x_3 + x_4 = 0$, or $x_2 + x_3 + x_4 = -2$. From (2), we know $x_2 + x_4 = 1$. Substituting this into our previous derived equation:

$$(x_2 + x_4) + x_3 = -2 \implies 1 + x_3 = -2 \implies x_3 = -3$$

Now substitute $x_1 = 2$ and $x_3 = -3$ into (1):

$$2(2) + x_2 + 2(-3) = 1 \implies 4 + x_2 - 6 = 1 \implies x_2 = 3$$

Finally, check (5) using $x_3 = -3$:

$$-3 + x_4 = 1 \implies x_4 = 4$$

However, we must verify if these values satisfy constraint (2): $x_2 + x_4 = 3 + 4 = 7$. But constraint (2) states $x_2 + x_4 = 1$. Since $7 \neq 1$, the system is **inconsistent**.

Conclusion: The primal problem is **infeasible**. No solution exists because the vector b does not lie in the column space of A.

Mitigation: To resolve this, one could use a Least Squares approach to minimize $\|Ax - b\|_2^2$, or relax the equality constraints to inequalities ($Ax \leq b$).

Comparison of Solutions

- **Primal:** Since the constraints are inconsistent, the primal optimal value is $p^* = \infty$.
- **Dual:** In the dual, if the primal is infeasible, the dual is typically unbounded ($d^* = \infty$) or infeasible.
- **Difference:** The difference arises from the lack of a feasible intersection of the hyperplanes defined by $Ax = b$. While the inequality version of this problem had a trivial solution at the origin, the equality version is more restrictive and fails to find any valid point in $\mathbb{R}^4$ that satisfies the geometry of A.

Solution for II.2

II.2 (i)

$$\text{rank}(A^\top A) = \text{rank}(A) \leq m < n$$

This implies the null space of A is not empty. So $A^\top A$ is an $n \times n$ matrix with rank $< n$ so it is not invertible, rather it is singular.

II.2 (ii)

Primal problem

$$\min_{x \in \mathbb{R}^n} ||x - x_0||_2^2 \quad \text{s.t. } Ax = y$$

Lagrangian

Let $\lambda \in \mathbb{R}^m$

$$L(x, \lambda) = ||x - x_0||_2^2 + \lambda^\top (Ax - y)$$

This problem has a quadratic objective and affine equality constraints $Ax = y$ so it is a convex optimization problem. We can assume $Ax = y$ is feasible so the problem is feasible as well. So Slater's condition is met. Therefore, **strong duality** holds.

KKT conditions

Stationarity

$$\nabla_x L = 2(x - x_0) + A^\top \lambda = 0 \implies x = x_0 - \frac{1}{2} A^\top \lambda$$

Primal feasibility

$$Ax = y$$

$$A\left(x_0 - \frac{1}{2}A^\top\lambda\right) = y$$
$$Ax_0 - \frac{1}{2}AA^\top\lambda = y$$
$$AA^\top\lambda = 2(Ax_0 - y)$$

We get the following since $AA^\top$ is invertible

$$\lambda^* = 2(AA^\top)^{-1}(Ax_0 - y)$$

Dual problem

$$g(\lambda) = \lambda^\top(Ax_0 - y) - \frac{1}{4}\lambda^\top(AA^\top)\lambda$$

Dual is

$$\boxed{\max_{\lambda\in\mathbb{R}^m} \lambda^\top(Ax_0 - y) - \frac{1}{4}\lambda^\top(AA^\top)\lambda}$$

Optimal primal and dual objective
We know $x^* - x_0 = A^\top(AA^\top)^{-1}(y - Ax_0)$, so

$$p^* = ||x^* - x_0||_2^2 = (y - Ax_0)^\top(AA^\top)^{-1}(y - Ax_0)$$

By strong duality, $d^* = p^*$.

II.2 (iii)

Gradient descent is represented by

$$x_{t+1} = x_t - 2\eta_t A^\top(Ax_t - y)$$

We can see that

$$A^\top(Ax_t - y) = \sum_{i=1}^{m}(a_i^\top x_t - y_i)a_i \in \text{span}\{a_i\}$$

So we can rewrite the update as

$$x_{t+1} = x_t + \left(-2\eta_t \sum_{i=1}^{m}(a_i^\top x_t - y_i)a_i\right)$$

By induction, our base case is $x_0 \in Q$. For each step, we can see that if $x_t \in Q$, then $x_{t+1} = x_t + v$ where $v \in \text{span}\{a_i\}$. Thus, all steps taken result in the iterates staying in Q.

II.2 (iv)

$$Ax - y = A(x_0 + A^\top u) - y = (Ax_0 - y) + AA^\top u$$

$$\min_u ||Ax_0 - y + AA^\top u||_2^2$$

$$Ax_0 - y + AA^\top u = 0 \implies u^* = (AA^\top)^{-1}(y - Ax_0)$$

Thus,

$$x^* = x_0 + A^\top u^* = x_0 + A^\top(AA^\top)^{-1}(y - Ax_0)$$

This is the same solution as the one from part (ii). Since $Ax = y$ is feasible and x^* satisfies it, the minimum objective value is 0.

II.2 (v)

We can first compute $r = y - Ax_0$.

$$Ax_0 = \begin{bmatrix} 8 \\ 4 \\ 17 \\ 12 \end{bmatrix}, \quad r = y - Ax_0 = \begin{bmatrix} -7 \\ -1 \\ -13 \\ -13 \end{bmatrix}$$

Then we can evaluate our $AA^\top$.

$$AA^\top = \begin{bmatrix} 4 & 3 & 5 & 3 \\ 3 & 6 & 5 & 1 \\ 5 & 5 & 18 & 3 \\ 3 & 1 & 3 & 11 \end{bmatrix}$$

Now, we solve $(AA^\top)u = r$.

$$\begin{bmatrix} 4 & 3 & 5 & 3 \\ 3 & 6 & 5 & 1 \\ 5 & 5 & 18 & 3 \\ 3 & 1 & 3 & 11 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} -7 \\ -1 \\ -13 \\ -13 \end{bmatrix}$$

We can do $R_2 \leftarrow 4R_2 - 3R_1, R_3 \leftarrow 4R_3 - 5R_1, R_4 \leftarrow 4R_4 - 3R_1$.

$$\begin{bmatrix} 4 & 3 & 5 & 3 & | & -7 \\ 0 & 15 & 5 & -5 & | & 17 \\ 0 & 5 & 47 & -3 & | & -17 \\ 0 & -5 & -3 & 35 & | & -31 \end{bmatrix}$$

Now, we do $R_3 \leftarrow 15R_3 - 5R_2, R_4 \leftarrow 15R_4 + 5R_2$.

$$\begin{bmatrix} 4 & 3 & 5 & 3 & | & -7 \\ 0 & 15 & 5 & -5 & | & 17 \\ 0 & 0 & 680 & -20 & | & -340 \\ 0 & 0 & -20 & 500 & | & -380 \end{bmatrix}$$

Then, we can divide by 20 to get

$$\begin{aligned} 34u_3 - u_4 &= -17 \\ -u_3 + 25u_4 &= -19 \end{aligned}$$

$$\begin{aligned} -u_3 + 25(34u_3 + 17) &= -19 \\ 849u_3 &= -444 \\ u_3 &= -\frac{148}{283} \end{aligned}$$

So

$$u_4 = 34\left(-\frac{148}{283}\right) + 17 = -\frac{221}{283}$$

Using R_2, we get $15u_2 + 5u_3 - 5u_4 = 17$; then

$$\begin{aligned} 3u_2 + u_3 - u_4 &= \frac{17}{5} \\ u_3 - u_4 &= -\frac{148}{283} + \frac{221}{283} = \frac{73}{283} \end{aligned}$$

Using R_1, $4u_1 + 3u_2 + 5u_3 + 3u_4 = -7$, we get

$$u_1 = -\frac{1834}{1415}$$

So

$$u = \begin{bmatrix} -\frac{1834}{1415} \\ \frac{1482}{1415} \\ -\frac{148}{283} \\ -\frac{221}{283} \end{bmatrix}$$

Now, we can find $x^* = x_0 + A^\top u$

$$A^\top u = \begin{bmatrix} u_1 + 2u_2 \\ u_1 + u_3 - u_4 \\ u_2 + u_3 \\ u_1 + 3u_4 \\ u_1 + u_2 + 4u_3 + u_4 \end{bmatrix} = \begin{bmatrix} \frac{226}{283} \\ -\frac{1469}{1415} \\ \frac{742}{1415} \\ -\frac{5149}{1415} \\ -\frac{4417}{1415} \end{bmatrix}$$

Using the given x_0, we get

$$x^* = \begin{bmatrix} \frac{226}{283} \\ 1 - \frac{1469}{1415} \\ \frac{742}{1415} \\ 3 - \frac{5149}{1415} \\ 4 - \frac{4417}{1415} \end{bmatrix} = \begin{bmatrix} \frac{226}{283} \\ -\frac{54}{1415} \\ \frac{742}{1415} \\ -\frac{904}{1415} \\ \frac{1243}{1415} \end{bmatrix}$$

Solution for II.3

(i)

The optimization problem is:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & x^\top \Sigma x \\ \text{s.t.} \quad & \mu^\top x \geq \alpha \\ & \mathbf{1}^\top x = 1 \end{aligned}$$

where $\mathbf{1}$ is the vector of all ones.

The Lagrangian is:

$$L(x, \lambda, \nu) = x^\top \Sigma x + \lambda(\alpha - \mu^\top x) + \nu(1 - \mathbf{1}^\top x)$$

The KKT conditions for optimality are:

1. $\nabla_x L(x, \lambda, \nu) = 2\Sigma x - \lambda\mu - \nu\mathbf{1} = 0$
2. $\mu^\top x \geq \alpha, \quad \mathbf{1}^\top x = 1$
3. $\lambda \geq 0$
4. $\lambda(\alpha - \mu^\top x) = 0$

From the first condition we can solve for x:

$$2\Sigma x = \lambda\mu + \nu\mathbf{1} \implies x = \frac{1}{2}\Sigma^{-1}(\lambda\mu + \nu\mathbf{1})$$

Next we substitute x into the two primal feasibility equations:

$$\mathbf{1}^\top x = 1 \implies \frac{1}{2}\mathbf{1}^\top \Sigma^{-1}(\lambda\mu + \nu\mathbf{1}) = 1$$
$$\mu^\top x = \alpha \implies \frac{1}{2}\mu^\top \Sigma^{-1}(\lambda\mu + \nu\mathbf{1}) = \alpha$$

To simplify the linear system, let us define:

$$A = \mathbf{1}^\top \Sigma^{-1}\mathbf{1}, \quad B = \mathbf{1}^\top \Sigma^{-1}\mu = \mu^\top \Sigma^{-1}\mathbf{1}, \quad C = \mu^\top \Sigma^{-1}\mu$$

Substituting these constants yields a system of two linear equations for the multipliers λ and ν:

$$B\lambda + A\nu = 2$$
$$C\lambda + B\nu = 2\alpha$$

In matrix form, this is:

$$\begin{pmatrix} C & B \\ B & A \end{pmatrix} \begin{pmatrix} \lambda \\ \nu \end{pmatrix} = \begin{pmatrix} 2\alpha \\ 2 \end{pmatrix}$$

The determinant of this matrix is $D = AC - B^2$. By the Cauchy-Schwarz inequality, $D > 0$. Inverting the matrix to solve for λ and ν:

$$\begin{pmatrix} \lambda \\ \nu \end{pmatrix} = \frac{1}{AC - B^2} \begin{pmatrix} A & -B \\ -B & C \end{pmatrix} \begin{pmatrix} 2\alpha \\ 2 \end{pmatrix} = \frac{2}{AC - B^2} \begin{pmatrix} A\alpha - B \\ C - B\alpha \end{pmatrix}$$

Thus, we have:

$$\lambda = \frac{2(A\alpha - B)}{AC - B^2}, \quad \nu = \frac{2(C - B\alpha)}{AC - B^2}$$

Finally, we substitute these optimal multipliers back into our earlier expression for x:

$$x^\star = \frac{1}{2}\Sigma^{-1}\left(\frac{2(A\alpha - B)}{AC - B^2}\mu + \frac{2(C - B\alpha)}{AC - B^2}\mathbf{1}\right)$$
$$x^\star = \frac{1}{AC - B^2}\Sigma^{-1}\big((A\alpha - B)\mu + (C - B\alpha)\mathbf{1}\big)$$

This is the exact closed-form primal solution in terms of Σ, α, μ, and the defined scalar constants A, B, and C.

(ii)

We do not know μ exactly. We are given $\hat{\mu}$ and uncertainty set

$$(\hat{\mu} - \mu)^\top A(\hat{\mu} - \mu) \le \varepsilon^2, \qquad A \in \mathbb{S}^n_{++},\ \varepsilon > 0.$$

(a)

$$\begin{array}{ll} \text{minimize} & \mu^\top x \\ \text{subject to} & (\hat{\mu} - \mu)^\top A(\hat{\mu} - \mu) \le \varepsilon^2. \end{array}$$

Change of variables. Let

$$z := A^{1/2}(\hat{\mu} - \mu) \quad \iff \quad \mu = \hat{\mu} - A^{-1/2}z.$$

Then the constraint becomes

$$\|z\|_2^2 \le \varepsilon^2 \quad \iff \quad \|z\|_2 \le \varepsilon.$$

Since $A \in \mathbb{S}^n_{++}$, $A^{1/2}$ exists and is invertible. The objective becomes

$$\mu^\top x = (\hat{\mu} - A^{-1/2}z)^\top x = \hat{\mu}^\top x - z^\top A^{-1/2}x.$$

Minimization. Since $\hat{\mu}^\top x$ is constant in z, minimizing $\hat{\mu}^\top x - z^\top A^{-1/2}x$ is equivalent to maximizing $z^\top A^{-1/2}x$ over $\|z\|_2 \le \varepsilon$:

$$\min_{\|z\|_2 \le \varepsilon} \left(\hat{\mu}^\top x - z^\top A^{-1/2}x\right) = \hat{\mu}^\top x - \max_{\|z\|_2 \le \varepsilon} z^\top A^{-1/2}x.$$

By Cauchy–Schwarz,

$$z^\top A^{-1/2}x \le \|z\|_2 \|A^{-1/2}x\|_2 \le \varepsilon \|A^{-1/2}x\|_2,$$

and equality is achieved by choosing

$$z = \varepsilon \frac{A^{-1/2}x}{\|A^{-1/2}x\|_2} \quad (\text{if } A^{-1/2}x \neq 0).$$

Hence

$$\min_{\mu:\ (\hat{\mu}-\mu)^\top A(\hat{\mu}-\mu) \le \varepsilon^2} \mu^\top x = \hat{\mu}^\top x - \varepsilon \|A^{-1/2}x\|_2.$$

(b)

Using the worst-case return lower bound, the robust problem is

$$\begin{array}{ll} \text{minimize} & x^\top \Sigma x \\ \text{subject to} & \hat{\mu}^\top x - \varepsilon \|A^{-1/2}x\|_2 \ge \alpha, \\ & \mathbf{1}^\top x = 1. \end{array}$$

Rewrite the inequality constraint as

$$\alpha - \hat{\mu}^\top x + \varepsilon \|A^{-1/2}x\|_2 \le 0.$$

Lagrangian. Introduce $\lambda \ge 0$ for the inequality and $\nu \in \mathbb{R}$ for the equality:

$$L(x, \lambda, \nu) = x^\top \Sigma x + \lambda \left(\alpha - \hat{\mu}^\top x + \varepsilon \|A^{-1/2}x\|_2\right) + \nu(\mathbf{1}^\top x - 1).$$

Dual function.

$$g(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \nu).$$

Group the x-terms:

$$L(x, \lambda, \nu) = x^\top \Sigma x + (\nu \mathbf{1} - \lambda \hat{\mu})^\top x + \lambda \varepsilon \|A^{-1/2}x\|_2 + (\lambda \alpha - \nu).$$

Use the dual norm to rewrite the SOC term. For any vector w, we have the dual norm identity

$$\|w\|_2 = \max_{\|u\|_2 \le 1} u^\top w.$$

Apply this with $w = A^{-1/2}x$:

$$\|A^{-1/2}x\|_2 = \max_{\|u\|_2 \le 1} u^\top A^{-1/2}x.$$

Therefore

$$\lambda \varepsilon \|A^{-1/2}x\|_2 = \max_{\|u\|_2 \le 1} (\lambda \varepsilon u^\top A^{-1/2}x).$$

So the Lagrangian can be written as

$$L(x, \lambda, \nu) = \max_{\|u\|_2 \le 1} \left[x^\top \Sigma x + \left(\nu \mathbf{1} - \lambda \hat{\mu} + \lambda \varepsilon A^{-1/2}u\right)^\top x + (\lambda \alpha - \nu)\right].$$

We interchange $\inf_x$ and $\max_u$:

$$g(\lambda, \nu) = \max_{\|u\|_2 \le 1} \left\{\inf_x \left[x^\top \Sigma x + \left(\nu \mathbf{1} - \lambda \hat{\mu} + \lambda \varepsilon A^{-1/2}u\right)^\top x\right] + (\lambda \alpha - \nu)\right\}.$$

Compute the inner infimum over x. For fixed (λ, ν, u), define

$$q := \nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u.$$

Since $\Sigma \succ 0$, the function $x^\top \Sigma x + q^\top x$ is strictly convex and minimized at

$$2\Sigma x + q = 0 \quad \Longrightarrow \quad x^\star(\lambda, \nu, u) = -\frac{1}{2}\Sigma^{-1}q.$$

The minimum value is the standard quadratic completion:

$$\inf_x \ \left(x^\top \Sigma x + q^\top x\right) = -\frac{1}{4}q^\top \Sigma^{-1} q.$$

Therefore

$$g(\lambda, \nu) = \max_{\|u\|_2 \le 1} \left[-\frac{1}{4}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big)^\top \Sigma^{-1}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big) + \lambda\alpha - \nu \right].$$

Dual problem. The dual is

$$\begin{array}{ll} \text{maximize} & \displaystyle\max_{\|u\|_2 \le 1} \left[-\frac{1}{4}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big)^\top \Sigma^{-1}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big) + \lambda\alpha - \nu \right] \\ \text{subject to} & \lambda \ge 0, \ \nu \in \mathbb{R}. \end{array}$$

Introducing u explicitly Equivalently, treat u as a dual variable directly:

$$\begin{array}{ll} \text{maximize} & -\frac{1}{4}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big)^\top \Sigma^{-1}\big(\nu\mathbf{1} - \lambda\hat{\mu} + \lambda\varepsilon A^{-1/2}u\big) + \lambda\alpha - \nu \\ \text{subject to} & \lambda \ge 0, \ \|u\|_2 \le 1, \ \nu \in \mathbb{R}. \end{array}$$

(iii)

For part (ii)b), the robust portfolio problem is

$$\begin{array}{ll} \text{minimize} & x^\top \Sigma x \\ \text{subject to} & \hat{\mu}^\top x - \varepsilon\|A^{-1/2}x\|_2 \ge \alpha, \\ & \mathbf{1}^\top x = 1, \end{array}$$

where $\hat{\mu} \in \mathbb{R}^n$, $\Sigma \in \mathbb{S}^n_{++}$, $A \in \mathbb{S}^n_{++}$, $\alpha > 0$, and $\varepsilon > 0$.

Plugging in the given values. Here $n = 3$,

$$\hat{\mu} = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}, \qquad \Sigma = \operatorname{diag}(1, 2, 4), \qquad A = I_3, \qquad \alpha = 3, \qquad \varepsilon = 0.1.$$

Since $A = I_3$, we have $A^{-1/2} = I_3$, hence $\|A^{-1/2}x\|_2 = \|x\|_2$. Therefore the constraint becomes

$$(-1,\ 2,\ 5)\, x - 0.1\|x\|_2 \ge 3, \qquad \text{and} \qquad x_1 + x_2 + x_3 = 1.$$

The objective simplifies to

$$x^\top \Sigma x = x_1^2 + 2x_2^2 + 4x_3^2.$$

This is a convex quadratic objective with a second-order cone (SOC) constraint, so it can be solved directly using CVXPY.

Implementation (CVXPY). Figure 2 shows the CVXPY code used to solve the SOCP. In CVXPY, the objective is implemented via `quad_form(x, Sigma)` and the robust constraint via `mu_hat @ x - eps * norm(x,2) >= alpha`, together with the equality constraint `sum(x) == 1`.

Computed optimizer. The solver returns the optimal allocation shown in Figure 3 (answer screenshot). Rounded to four decimal places, the solution is

$$x^\star \approx \begin{bmatrix} 0.0906 \\ 0.4639 \\ 0.4455 \end{bmatrix}.$$

This allocation satisfies $\mathbf{1}^\top x^\star = 1$, and the robust return constraint $\hat{\mu}^\top x^\star - 0.1\|x^\star\|_2 \geq 3$ is active up to numerical tolerance (as expected at optimum).

Screenshots. The CVXPY code and the solver output are included below.

```python
import cvxpy as cp
import numpy as np

# Data
mu_hat = np.array([-1.0, 2.0, 5.0])
Sigma  = np.diag([1.0, 2.0, 4.0])
A = np.identity(3)
alpha  = 3.0
eps    = 0.1
n      = 3

# Variable
x = cp.Variable(n)

# Objective: x^T Sigma x
objective = cp.Minimize(cp.quad_form(x, Sigma))

# Constraints: mu_hat^T x - eps*||x||_2 >= alpha, 1^T x = 1 (A^1/2 = I)
constraints = [
    mu_hat @ x - eps * cp.norm(x, 2) >= alpha,
    cp.sum(x) == 1
]

prob = cp.Problem(objective, constraints)

prob.solve(solver=cp.SCS)

print("status:", prob.status)
print("optimal value:", prob.value)
print("x*:", x.value)
print("x* rounded:", np.round(x.value, 4))
```

Figure 2: CVXPY implementation for solving the robust SOCP in part (ii)(b).

```
(base) → Convex Optimization python3 hw4.py
status: optimal
optimal value: 1.2326240286746408
x*: [0.09056547 0.46389331 0.44554122]
x* rounded: [0.0906 0.4639 0.4455]
```

Figure 3: Solver output and computed optimal allocation $x^\star$ (rounded to four decimals).